Tanvir Ahmed Fahim

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Research Interests

- RF and microwave engineering
- Applied and computational electromagnetics
- Antenna design and equivalent-circuit modeling
- mmWave technologies for 5G communications
- Next-generation 6G wireless systems
- THz devices and high-speed wireless applications
- IoT-enabled wireless systems
- Machine learning-assisted antenna optimization

Education

- **Bangladesh Army International University of Science and Technology** Cumilla, Bangladesh
  Bachelor of Science in Electrical and Electronic Engineering (EEE) *Oct 2022 – Oct 2026*
  Emphasis: RF & Microwave, Antenna Design, Applied Electromagnetics, and Wireless Communication
  Current CGPA: 3.81/4.00 (Thesis defense pending)
  Thesis Title: A Compact Butterfly-Shaped Slot Microstrip MIMO Antenna for High-Performance Wideband THz Applications

Work Experience

- **Nagorik TV** Dhaka, Bangladesh
  Designation: Industrial Trainee *03 Sep 2026 – 24 Sep 2026*
  - Gained practical exposure to television broadcasting operations, including studio production, control-room workflows, content management, and transmission systems.
  - Developed an understanding of the complete broadcast workflow across Studio, PCR, MCR, Ingest, Editing, Newsroom, CER, Archive, and RF transmission.
- **Bangladesh Army** Bangladesh
  Designation: Soldier *Jul 2019 – Apr 2022*
  - Served in signals units to support military communication networks and tactical field operations, strengthening discipline and teamwork across multiple operational roles.
- **Field Intelligence Unit** Cox's Bazar, Bangladesh
  Designation: Wireless Operator *Oct 2020 – May 2021*
  - Served in the Field Intelligence Unit of the Bangladesh Army as a Wireless Operator, operating secure communication systems and supporting intelligence missions.

Research Projects

- **ESP32-Based Multimodal Health Monitoring System** Jul 2026
  ESP32, Biomedical Instrumentation, IoT, MAX30100, EEG, Blynk
  - Designed and developed an ESP32-based multimodal health monitoring system integrating a MAX30100 pulse-oximeter, three-electrode EEG sensor, and OLED display for real-time physiological-signal acquisition, with Wi-Fi-enabled remote monitoring through the Blynk IoT platform.
- **Advanced Pipe Inspection Robot** Jan 2025
  Control Systems, Navigation, Sensors, Real-Time Data Transmission
  - Designed an advanced pipe inspection robot for complex pipeline infrastructures, featuring a wall-press mechanism, a sensor suite, an adaptive control system, and real-time data transmission.
- **Smart Car Parking System** Dec 2024
  IoT, Microcontroller, System Design and Setup
  - Developed an IoT-based smart parking system utilizing IR obstacle sensors, servo motors, and LED indicators for automated gate control and real-time occupancy monitoring.

- **IoT-Based Transmission Line Fault Detection System for Bangladesh's Power Grid** Oct 2024
  GPS, GSM, Fault Detection, Transmission Line
  - Designed an IoT-enabled system for detecting and locating transmission line faults using real-time sensor data, GPS for fault localization, GSM for instant alerts, and Wi-Fi for remote monitoring.
- **Design and Implementation of FIR Filters for Image Processing Applications** Aug 2024
  Digital Signal Processing, MATLAB, FIR Filter Design
  - Implemented FIR filter design techniques for edge detection and noise reduction in image processing applications.

Published and Accepted Publications

1. **T. A. Fahim**, M. A. A. Mamun, J. K. Dea, and M. G. Mustafa, "Machine Learning-Driven Gain Prediction of a High-Performance Dual-Wideband Decoupling-Enhanced Graphene MIMO Antenna for THz Applications," *Diamond & Related Materials*. DOI: 10.1016/j.diamond.2026.114121.
2. M. N. Alam, M. Z. H. Buiyan, M. A. A. H. Anik, F. Tanvir, and **T. A. Fahim**, "Enhancing Power Grid Reliability in Bangladesh: An IoT-Based Approach to Transmission Line Fault Detection," in *2024 IEEE 3rd International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON)*, Dhaka, Bangladesh: IEEE, Nov. 2024, pp. 224–229. DOI: 10.1109/RAAICON64172.2024.10928383.
3. **T. A. Fahim**, M. S. Rana, N. M. M. Mujahid, and T. Hasan, "28 GHz Patch Antennas for 5G Wireless Applications: A Comprehensive Review," in *Information Systems Engineering and Management*, 2026, pp. 205–218. DOI: 10.1007/978-3-032-19196-0_16.
4. **T. A. Fahim**, M. A. A. Mamun, S. A. Bipul, T. Hasan, and H. Datta, "Design of a High Performance Nano Band Microstrip Antenna with Three Rectangular Side Slots for THz Applications," *IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN 2026)*. DOI: 10.1109/QPAIN69676.2026.11545910.
5. **T. A. Fahim**, M. S. Rana, S. A. Bipul, T. Hasan, N. M. M. Mujahid, and H. Datta, "Design of a Compact Modified Inset-Fed Circular and Inverted U-Shaped Patch Antenna for Terahertz Applications," *11th International Conference on Information and Communication Technology for Intelligent Systems (ICTIS 2026)*. **(Accepted)**
6. M. A. Haque, **T. A. Fahim**, M. H. Riad, J. J. Tiang, and N. S. S. Singh, "Compact THz MIMO Antenna with Multi-Resonance and Low Mutual Coupling for High-Speed Communication," *Engineering, Technology & Applied Science Research*. **(Accepted)**
7. **T. A. Fahim**, M. A. A. Mamun, J. K. Dea, and N. J. Sumaiya, "A Dual-Band High-Gain Multiresonance Graphene Based THz Antenna for Future 6G Applications," *2026 IEEE International Conference on Microwave, Antennas, RF and Sensors*. **(Accepted)**
8. **T. A. Fahim**, M. A. A. Mamun, M. G. Mustafa, and M. A. Haque, "Isolation-Wall Integrated Graphene THz MIMO Antenna with Wideband and High-Diversity Performance for 6G Applications," *2nd International Conference on Advances in Computing, Communication, Electrical, and Smart Systems (iCACCESS 2026)*. **(Accepted)**

Publications (Under Review)

1. **T. A. Fahim**, M. H. Riad, J. J. Tiang, N. S. S. Singh, and M. A. Haque, "Design and Performance Analysis of a PBG Substrate MIMO Antenna with RLC Validation for 6G and IoT Applications," *Discover Internet of Things*.
2. **T. A. Fahim**, M. H. Riad, J. J. Tiang, N. S. S. Singh, and M. A. Haque, "A Novel Clustered-Slot THz Multi-Element Radiator with Circuit Validation and Hybrid Learning for UAV-Assisted 6G Smart Cities," *IEEE Open Journal of the Communications Society*.
3. M. A. Haque, **T. A. Fahim**, M. H. Riad, J. J. Tiang, N. S. S. Singh, Md. M. Islam, M. R. Rashel, and M. Tlemcani, "ML/ DL Assisted Design of a Flower-Shaped Two-Port Antenna for 6G IoT Applications with ECM Validation," *Engineering Reports*.

4. **T. A. Fahim**, M. A. A. Mamun, J. K. Dea, and M. G. Mustafa, "Compact Multiresonant Graphene-Based THz Antenna with RLC Circuit Validation and ML-Assisted Efficiency Prediction for Advanced Wireless Applications," *Scientific Reports*.
5. M. A. Haque, **T. A. Fahim**, M. H. Riad, J.-J. Tiang, and N. S. S. Singh, "Design and Multi-Level Validation of a Penta-Slot Graphene THz MIMO Antenna Using RLC Circuit Modeling and Hybrid Machine Learning," *SN Computer Science*.
6. M. A. Haque, **T. A. Fahim**, M. H. Riad, Md. S. Mahmud, J. J. Tiang, and N. S. S. Singh, "High-Performance Quad-Resonant THz MIMO Antenna with Twin Decoupling Walls for Short-Range Industrial IoT Applications," *Engineering, Technology & Applied Science Research*.
7. **T. A. Fahim**, M. G. Mustafa, M. S. Uddin, J. K. Dea, and K. R. U. Rahman, "Super-Wideband Graphene-Based THz MIMO Antenna with Machine-Learning-Based Performance Prediction for IoT- and IoNT-Enabled 6G Smart Cities," *Measurement*.

Achievements

- **Best Poster Award, BAIUST Inter Department Poster Presentation Competition** *Dec 2025*
- **10%–40% Tuition Fee Waiver for VC List Recognition (1st–4th, 6th–7th semesters)** *2022–2026*
- **Best Soldier Award – 2021 (9 Signals Battalion)** *Dec 2021*
- **Board Scholarship of Secondary School Certificate Examination** *Feb 2016*

Languages and Technical Skills

Languages: Bengali (Native), English (Proficient)
Simulation & EM Tools: HFSS, CST, COMSOL, ADS
Programming: Python, C, Arduino, MATLAB
Circuit & VLSI Design: LTspice, PSpice, Proteus, PSCAD, Microwind
3D Design: SolidWorks, AutoCAD
Other Tools: Simulink, Visio, LaTeX, Origin, SigmaPlot

Community Service & Volunteering Activities

- **IEEE BAIUST Student Branch — Senior Executive Member** *Nov 2024 – Nov 2025*
- **BAIUST Robotics Society — General Secretary** *Sep 2024 – Sep 2025*
- **Bangladesh National Cadet Corps — Cadet** *Jun 2016 – Dec 2017*
- **Shaheen Academy School and College Scouts — Member** *Jan 2011 – Jan 2016*

Referees

Dr. Mohammad Shafiul Alam
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